

Solution: This statement is True. It is mentioned on page 59 just above the section "Sets of Two or More Vectors".

b) If a set contains fewer vectors than there are entries in the vectors, then the set is linearly independent.

Solution: Let,  $u = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$  and  $v = \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}$ . We can easily show that  $v$  is a scalar multiple of  $u$ , where  $v = \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} = 2u$

On page 59 inside the first grey box it's mentioned that a set of two vectors  $\{v_1, v_2\}$  is linearly dependent if at least one of the vectors is a multiple of the other.  $\{u, v\}$  are a set of two vectors having 3 entries each. They are linearly dependent.  $\therefore$  The given statement is False.  $\square$

c) If  $x$  and  $y$  are linearly independent, and if  $z$  is in  $\text{Span}\{x, y\}$ , then  $\{x, y, z\}$  is linearly dependent.

Solution: By the defn. of Linear Independence on page 57, if  $c_1x + c_2y = 0$  for some  $c_1, c_2 \in \mathbb{R}$ , then  $c_1 = c_2 = 0$ . By the defn. of Span on page 30, for some  $p, q \in \mathbb{R}$ ,  $px + qy = z \Rightarrow px + qy - 1 \cdot z = 0$ . From the defn. of linear dependence on page 57,  $\{x, y, z\}$  are linearly dependent.  $\square$

$\therefore$  The given statement is True.

d) If a set in  $\mathbb{R}^n$  is linearly dependent, then the set contains more vectors than there are entries in each vector.

Solution: This statement is False. The solution is similar to Prob 22(b).  $\square$

3.  $\begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix}, \begin{bmatrix} -2 \\ -9 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ h \\ -9 \end{bmatrix}$  Solution: By defn. of linear dependence,  $\exists c_1, c_2, c_3 \in \mathbb{R}$  s.t.  $c_1 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ -9 \\ 6 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ h \\ -9 \end{bmatrix} = 0$  and not all of  $c_1, c_2$  and  $c_3$  are 0 together.

By defn. of matrix equation  $Ax = b$ , we have:  $\begin{bmatrix} 1 & -2 & 3 \\ 5 & -9 & h \\ -3 & 6 & -9 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$